

Conversational AI — Group Assignment 1

Problem Statement – 2 : Study of Embedding Models and Approximate Nearest Neighbor Search — Semantic Quality vs Search Efficiency

General Instructions

1. Students must carefully read and follow the instructions provided in each problem statement and task.
2. Students must submit the completed assignment notebook in both formats:
 - a. Jupyter Notebook format (.ipynb)
 - b. PDF format of the executed notebook
3. The Jupyter Notebook must contain the output displayed for every executed cell.
4. Assignments developed or submitted using any Python IDE other than the prescribed environment will not be considered for grading.
5. Students must ensure that they submit the correct Assignment Set assigned to them. Submission of an incorrect Assignment Set will not be evaluated.
6. For each and every task, students must provide a detailed explanation, justification, and inference. Merely providing code and output will not be sufficient for evaluation.
7. The notebook should clearly include:
 - a. Assignment title
 - b. Student details
 - c. Contribution by each student, including the section that has been done.
 - d. Problem statement
 - e. Dataset details and source
 - f. Tools and libraries used
 - g. Code implementation
 - h. Output screenshots/results
 - i. Explanation of the logic used
 - j. Justification for the chosen method/model/approach
 - k. Inference drawn from the results
 - l. Limitations observed
 - m. Possible improvements
 - n. Final conclusion
 - o. References
8. Students must clearly explain the architectural design, workflow, and reasoning behind the selected approach. For RAG-based assignments, this includes chunking, retrieval strategy, ranking/re-ranking, context construction, and evaluation. For fine-tuning-based assignments, this includes dataset preparation, training strategy, model adaptation method, preference optimization, and validation.
9. Students must not simply use built-in APIs or pre-trained pipelines without explaining the underlying process. The implementation should clearly demonstrate conceptual understanding of the selected method.
10. Incomplete submissions, missing outputs, unclear explanations, absence of either the .ipynb file or the final PDF report, missing Virtual Lab screenshots, or lack of proper justification and inference may lead to a reduction of marks.

11. Students must prepare a final report in PDF format and upload it along with the .ipynb notebook. If the final PDF report is not attached, the submission will be considered partial, and marks will be reduced accordingly.
12. Students are advised to verify the complete notebook execution from start to end before submission. Notebooks with execution errors, missing cells, broken links, or unavailable datasets may be penalized.
13. Any copied content, direct use of AI-generated answers without proper understanding, or submission without meaningful explanation and inference may lead to mark deduction.
14. Proper references must be provided for datasets, models, libraries, research papers, APIs, documentation, or external resources used in the assignment.
15. Students must ensure that all figures, tables, architecture diagrams, workflow diagrams, and evaluation results are properly labelled and explained.
16. The final conclusion must summarize the key observations, strengths of the implemented approach, limitations, and possible future improvements.

Group Assignment Contribution Guidelines

This is a **group assignment**. Each team must clearly document the contribution of each member.

Contribution Area	Description
Dataset & embedding	Prepare the dataset with the queries and generate the embeddings
Similarity Matrix comparison	Comparison of the embedding models
KNN Baseline	Knn implementation for each baseline
HNSW vs IVF	Comparison study between HNSW & IVF & ANN tradeoff
Evaluation & Analysis	Metrics computation, analysis, and visualization
Visualization & Reporting	All visualizations, architecture diagrams, final report

Each member must contribute to **at least one area**. The contribution of each member must be clearly stated in the notebook and final report.

Total: 10 Marks

Problem Statement

Module 1: Dataset and Embedding Preparation — 1 Marks

Task 1: Corpus and Query Dataset Preparation — 0.5 Marks

Select one publicly available text dataset suitable for semantic retrieval. And create:

- A document corpus of **at least 1,000 text passages**
- A query set of **at least 50 queries**
- Relevance information for each query

Students may use an existing dataset containing query-document relevance labels.

Task 2: Embedding Generation and Pooling — 0.5 Marks

Select two accessible encoder-based embedding models. Generate embeddings for the same corpus using both models.

For each model, document:

- Model name
- Embedding dimension
- Maximum/typical input length
- Pooling strategy used
- Approximate time required to generate the embeddings

Explain why encoder models are appropriate for semantic representation and retrieval. Do this using at least two different models and make a comparison study.

Module 2: Similarity and Exact Retrieval — 3 Marks

Task 3: Similarity Metric Comparison — 1.5 Marks

1. For one selected embedding model, compare the following similarity measures:
 - Cosine similarity
 - Dot product
 - L2/Euclidean distance
2. Select at least **20 query-document pairs** and calculate the similarity/distance values.
3. Analyse:
 1. Whether the ranking of documents changes across metrics
 2. Whether normalization affects the results
 3. Which metric appears most suitable for your selected embeddings

The analysis should refer to the mathematical definition of the similarity measures rather than simply reporting library output.

Task 4: Exact kNN Baseline — 1.5 Marks

1. Implement an exact nearest-neighbor search baseline.
2. For each query:
 - Generate the query embedding
 - Compare it against all document embeddings

- Retrieve Top-5 documents
3. Measure:
 - Search/query latency
 - Recall@5
 - Number of vectors examined per query
 4. Report average latency across the query set.

Module 3: ANN Search Experiment — 3 Marks

Task 5: HNSW vs IVF — 2 Marks

1. Implement at least **two ANN approaches**:

- HNSW
- IVF

2. Use the same embedding vectors and query set for both approaches.

- For HNSW, vary at least one relevant search parameter.
- For IVF, vary **nprobe** across at least three values.

3. Measure for each configuration:

- Recall@5
- Average query latency
- Search speed relative to exact search
- Number/fraction of candidate vectors examined, where measurable

Task 6: ANN Trade-off Analysis — 1 Marks

1. Create at least two plots:

1. **Recall@5 vs Query Latency**
2. **Recall@5 vs Search Efficiency**

2. Identify:

- Fastest configuration
- Highest-recall configuration
- Best balanced configuration

3. If this retrieval system had to serve a large number of queries, which ANN configuration would you select and why? Your answer must be supported by the experimental results.

Module 4: Embedding Quality Analysis — 3 Marks

Task 7: Qualitative Retrieval Analysis — 2 Marks

1. Select **10 representative queries** and compare the Top-5 results produced by:

- Embedding Model 1
- Embedding Model 2

2. Identify at least three types of retrieval differences, such as:

- Semantic similarity
- Keyword overlap
- Domain-specific terminology
- Ambiguous queries
- Short queries
- Long queries

3. Provide examples and explain why the models may produce different rankings.

Task 8: Final Recommendation — 1 Marks

Prepare a concise recommendation covering:

1. Which embedding model performed better?
2. Which similarity metric was most appropriate?
3. Which ANN method provided the best trade-off?
4. What configuration would you select for a resource-constrained deployment?
5. What would be the main limitation of your chosen approach?
6. How would you deploy one of the selected models in production?

Recommendations must be based on experimental evidence rather than general statements.

Deliverables

Each group must submit:

1. Python notebook (.ipynb)

2. PDF report, Maximum 30 pages

- Experimental methodology
- Comparative tables
- Graphs
- Retrieval examples
- Analysis and recommendations
- One-page executive summary

Submit all files as **one ZIP file per group**.